



BLM3620 Digital Signal Processing

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Course Materials



Important Materials:

- James H. McClellan, R. W. Schafer, M. A. Yoder, *DSP First Second Edition*, Pearson, 2015.
- Lizhe Tan, Jean Jiang, *Digital Signal Processing: Fundamentals and Applications*, Third Edition, Academic Press, 2019.

Auxiliary Materials:

- Prof. Sarp Ertürk, *Sayısal İşaret İşleme*, Birsen Yayınevi.
- Prof. Nizamettin Aydın, DSP Lecture Notes.
- J. G. Proakis, D. K. Manolakis, *Digital Signal Processing Fourth Edition*, Pearson, 2014.
- J. K. Perin, *Digital Signal Processing, Lecture Notes*, Stanford University, 2018.

Syllabus



Week	Lectures
1	Introduction to DSP and MATLAB
2	Sinuzoids and Complex Exponentials
3	Spectrum Representation
4	Sampling and Aliasing
5	Discrete Time Signal Properties and Convolution
6	Convolution and FIR Filters
7	Frequency Response of FIR Filters
8	Midterm Exam
9	Discrete Time Fourier Transform and Properties
10	Discrete Fourier Transform and Properties
11	Fast Fourier Transform and Windowing
12	z- Transforms
13	FIR Filter Design and Applications
14	IIR Filter Design and Applications
15	Final Exam

For more details -> Bologna page: <http://www.bologna.yildiz.edu.tr/index.php?r=course/view&id=5730&aid=3>

Lecture #8 – Frequency Response of FIR Filters

- Frequency Response
- Digital Filtering
- Frequency Scaling
- Exercises
- FIR Filter Application

Remember: Classification of Impulse Response $h[n]$

FIR – Finite Impulse Response:

- Number of impulses are limited.
- Always stable.

For example: $h[n] = \delta[n - 1] + 5\delta[n - 5]$

IIR – Infinite Impulse Response:

- Number of impulses are infinite.
- Sometimes these systems are not stable.

For example: $h[n] = u[n - 1] + 5u[n - 5]$

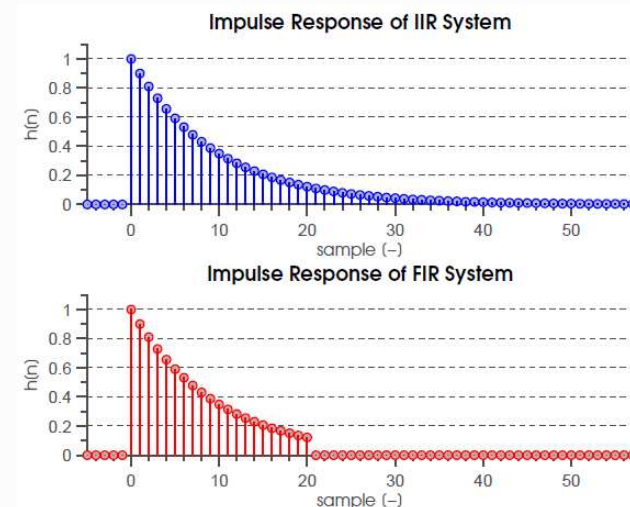
Finite impulse response (FIR):

$$y(n) = \sum_{k=0}^{M-1} h(k)x(n-k)$$

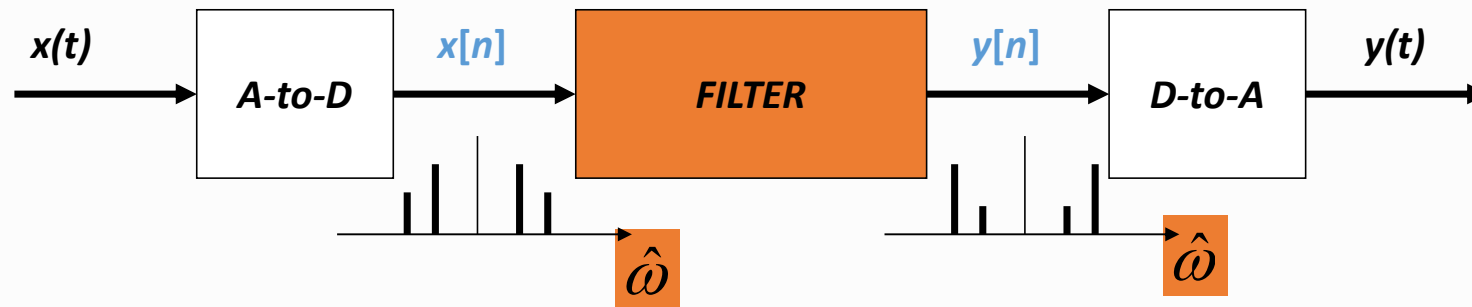
Infinite impulse response (IIR):

$$y(n) = \sum_{k=0}^{\infty} h(k)x(n-k)$$

Another example:



DOMAINS: Time & Frequency



- CONCENTRATE on the SPECTRUM
- SINUSOIDAL INPUT
 - INPUT $x[n]$ = SUM of SINUSOIDS
 - Then, OUTPUT $y[n]$ = SUM of SINUSOIDS
- Time-Domain: "n" = time
 - $x[n]$ discrete-time signal
 - $x(t)$ continuous-time signal

Example FIR Filters

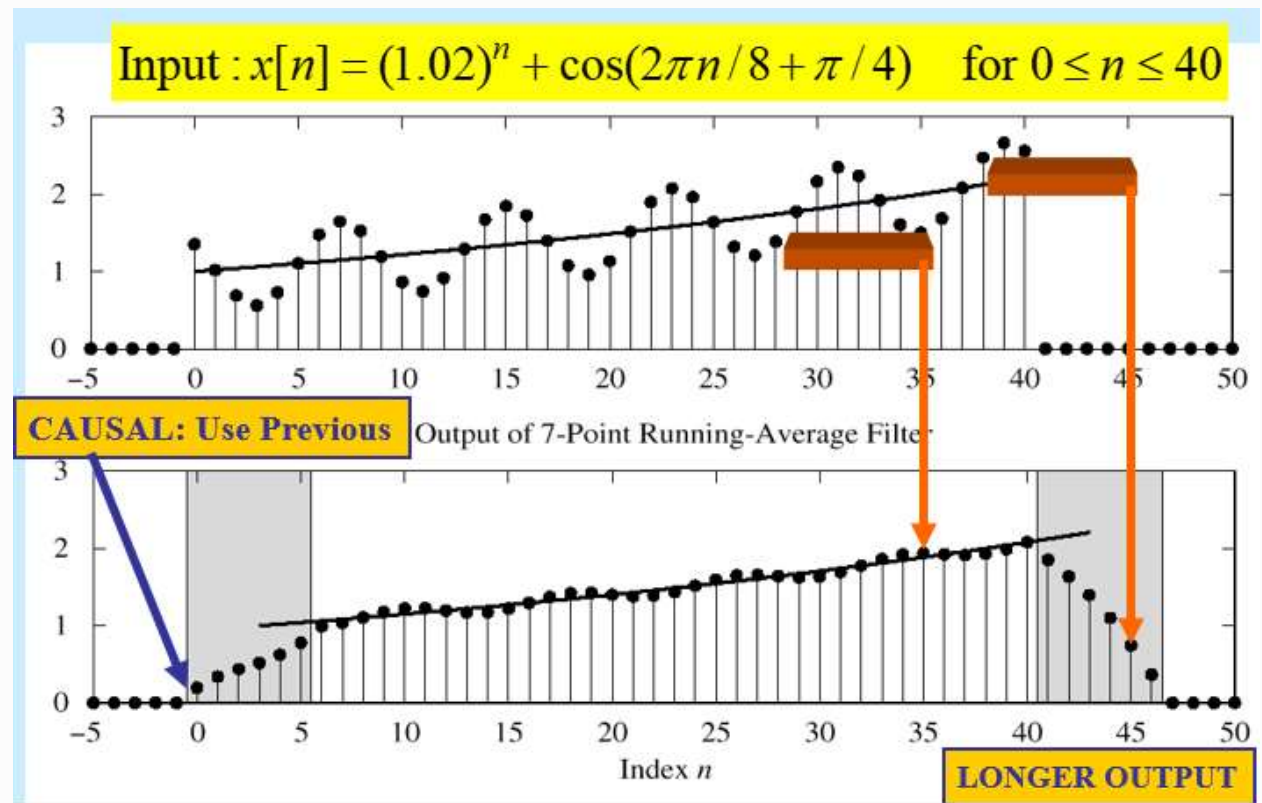


- 3-point AVERAGER

$$y_3[n] = \sum_{k=0}^2 \left(\frac{1}{3}\right) x[n-k]$$

- 7-point AVERAGER

$$y_7[n] = \sum_{k=0}^6 \left(\frac{1}{7}\right) x[n-k]$$



But...



How can I calculate the effects of this filter on digital frequency?

$$x[n] = Ae^{j\varphi} e^{j\hat{\omega}n} \quad -\infty < n < \infty$$

$x[n]$ is the input signal—a complex exponential

$$y[n] = \sum_{k=0}^M b_k x[n-k] = \sum_{k=0}^M h[k] x[n-k]$$

FIR DIFFERENCE EQUATION

$$H(e^{j\hat{\omega}}) = H(\hat{\omega}) = \sum_{k=-\infty}^{\infty} h[k] \cdot e^{-j\hat{\omega}k}$$

- Use the FIR “Difference Equation”

$$y[n] = \sum_{k=0}^M b_k x[n-k] = \sum_{k=0}^M b_k Ae^{j\varphi} e^{j\hat{\omega}(n-k)}$$

$$= \left(\sum_{k=0}^M b_k e^{j\hat{\omega}(-k)} \right) Ae^{j\varphi} e^{j\hat{\omega}n}$$

$$= H(\hat{\omega}) Ae^{j\varphi} e^{j\hat{\omega}n}$$

New Term: Frequency Response $H(e^{j\hat{\omega}})$

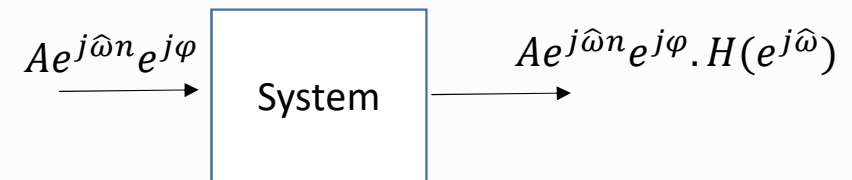
- At each frequency, we can **DEFINE**

$$H(e^{j\hat{\omega}}) = \sum_{k=0}^M b_k e^{-j\hat{\omega}k} \quad \leftarrow \text{FREQUENCY RESPONSE}$$

- Complex-valued formula
 - Has **MAGNITUDE** vs. frequency
 - And **PHASE** vs. frequency
- Notation: $H(e^{j\hat{\omega}})$ in place of $H(\hat{\omega})$

$$\begin{aligned} H(e^{j\hat{\omega}}) &= h[0] + h[1]e^{-j\hat{\omega}} + h[2]e^{-j2\hat{\omega}} + h[3]e^{-j3\hat{\omega}} + \dots \\ &= |H(e^{j\hat{\omega}})| e^{j\angle H(e^{j\hat{\omega}})} \end{aligned}$$

$|H(e^{j\hat{\omega}})| \rightarrow$ even symmetric
 $\angle H(e^{j\hat{\omega}}) \rightarrow$ odd " "



Complex Number:

- 1- A Phase Component
- 2- A Magnitude Component

$$H e^{j\omega} = \sum_{k=0}^{\infty} h[k] \cdot e^{-j\omega k}$$

$$H(e^{j\omega}) = \sum_{k=0}^{\infty} h[k] \cdot e^{-j\omega k} = 1 + 2 \cdot e^{-j\omega} + 1 \cdot e^{-j\omega 2}$$

Example

$$H(e^{j\hat{\omega}}) = \sum_{k=0}^M b_k e^{-j\hat{\omega} k}$$

not $\rightarrow \omega$ old ω
i.e. $-\pi - \pi$ aralarında
diger m. r.
cünki
digital
frequencies

$$\{b_k\} = \{1, 2, 1\}$$

**EXPLOIT
SYMMETRY**

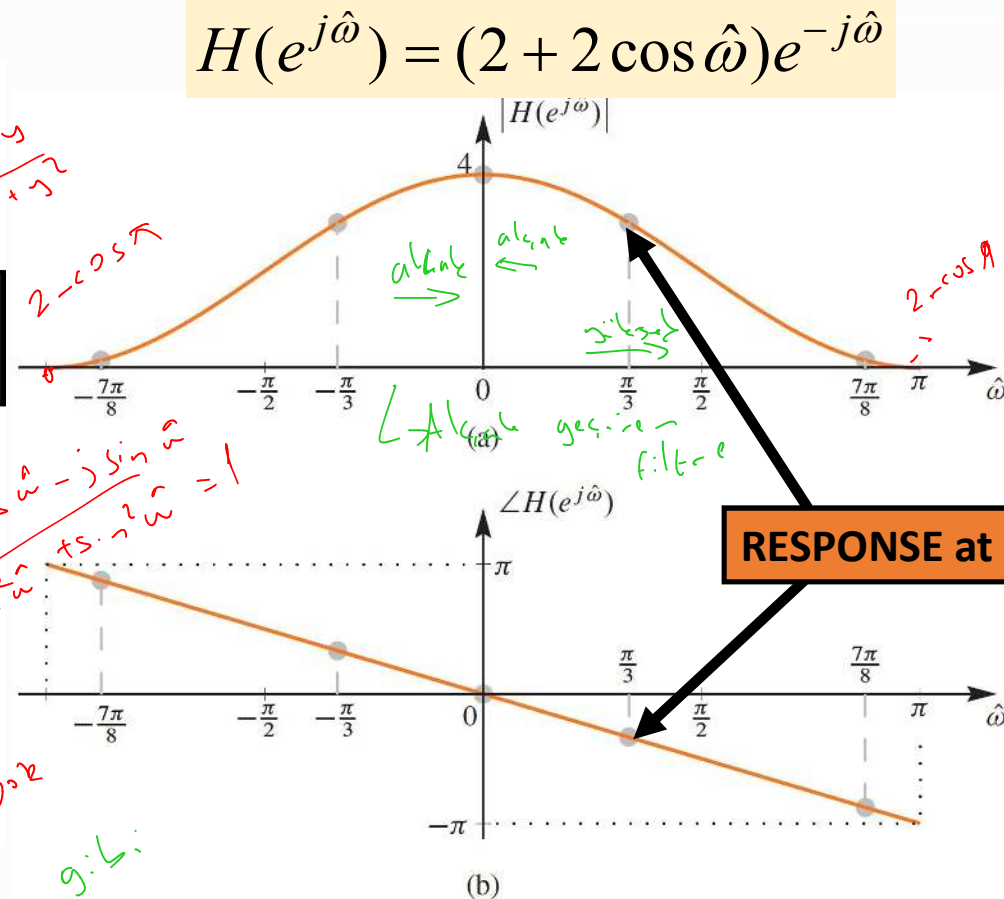
$$\begin{aligned} H(e^{j\hat{\omega}}) &= 1 + 2e^{-j\hat{\omega}} + e^{-j2\hat{\omega}} \\ &= e^{-j\hat{\omega}} (e^{j\hat{\omega}} + 2 + e^{-j\hat{\omega}}) \\ &= e^{-j\hat{\omega}} (2 + 2\cos\hat{\omega}) \end{aligned}$$

Since $(2 + 2\cos\hat{\omega}) \geq 0$

Magnitude is $|H(e^{j\hat{\omega}})| = (2 + 2\cos\hat{\omega})$

and Phase is $\angle H(e^{j\hat{\omega}}) = -\hat{\omega}$

What does this filter do in frequency domain?



Example – 2 : For the previous system...

Find $y[n]$ when $x[n] = 2e^{j\pi/4} e^{j(\pi/3)n}$

One Step - evaluate $H(e^{j\hat{\omega}})$ at $\hat{\omega} = \pi/3$

$$H(e^{j\hat{\omega}}) = (2 + 2 \cos \hat{\omega}) e^{-j\hat{\omega}}$$

$$H(e^{j\hat{\omega}}) = 3e^{-j\pi/3} \quad @ \hat{\omega} = \pi/3$$

$$y[n] = (3e^{-j\pi/3}) \times 2e^{j\pi/4} e^{j(\pi/3)n} = 6e^{-j\pi/12} e^{j(\pi/3)n}$$

Example - 3 : For the previous system...

Find $y[n]$ when $x[n] = 2\cos(\frac{\pi}{3}n + \frac{\pi}{4})$

$$2\cos(\frac{\pi}{3}n + \frac{\pi}{4}) = e^{j(\pi n/3 + \pi/4)} + e^{-j(\pi n/3 + \pi/4)}$$

$$\Rightarrow x[n] = x_1[n] + x_2[n]$$

Use
Linearity

$$y_1[n] = H(e^{j\pi/3})e^{j(\pi n/3 + \pi/4)}$$

$$y_2[n] = H(e^{-j\pi/3})e^{-j(\pi n/3 + \pi/4)}$$

$$\Rightarrow y[n] = y_1[n] + y_2[n]$$

$e^{-j\omega} \cdot (2 \cdot \cos \omega n)$

$e^{j\omega n} + e^{-j\omega n}$

$e^{j\omega n} + e^{-j\omega n}$

$e^{j\omega n} + e^{-j\omega n}$

Example - 3 : For the previous system...

Find $y[n]$ when $x[n] = 2 \cos(\frac{\pi}{3}n + \frac{\pi}{4})$

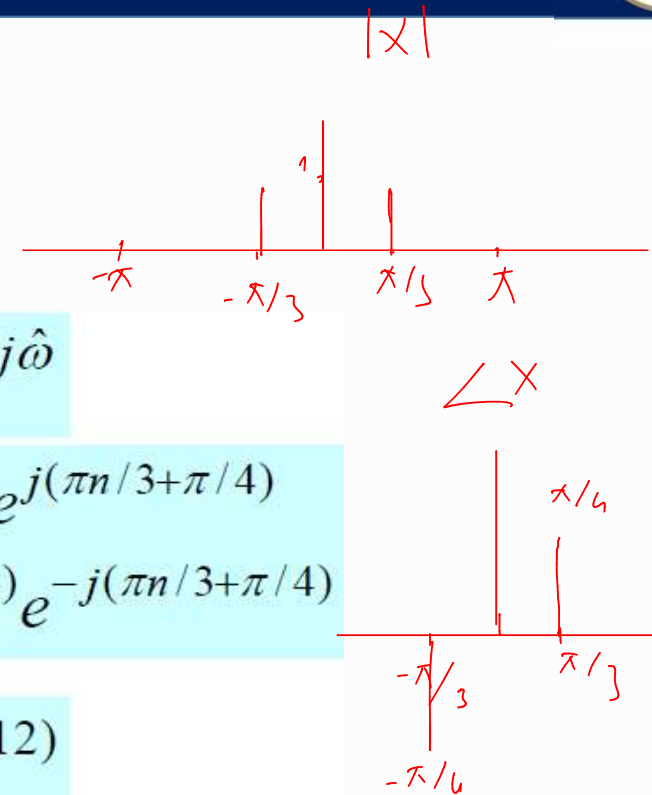
$$H(e^{j\hat{\omega}}) = (2 + 2 \cos \hat{\omega})e^{-j\hat{\omega}}$$

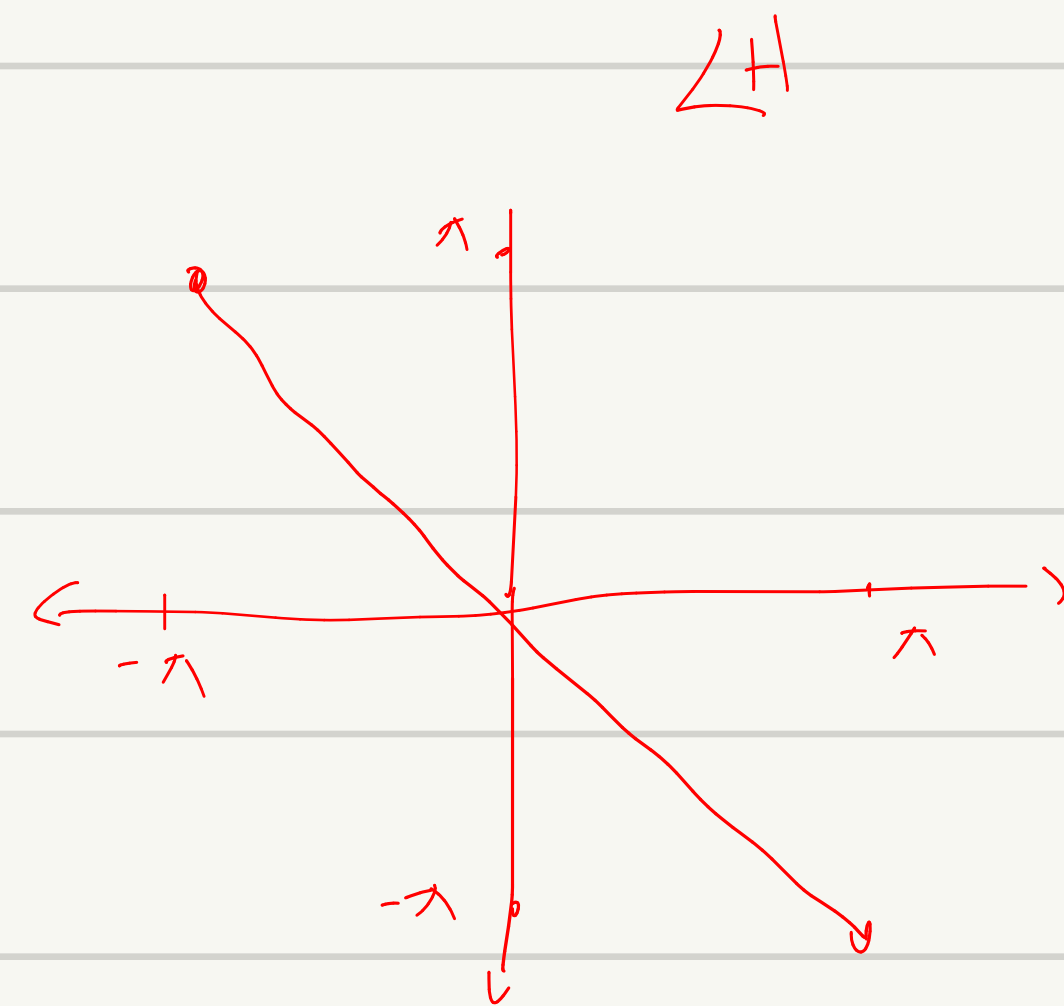
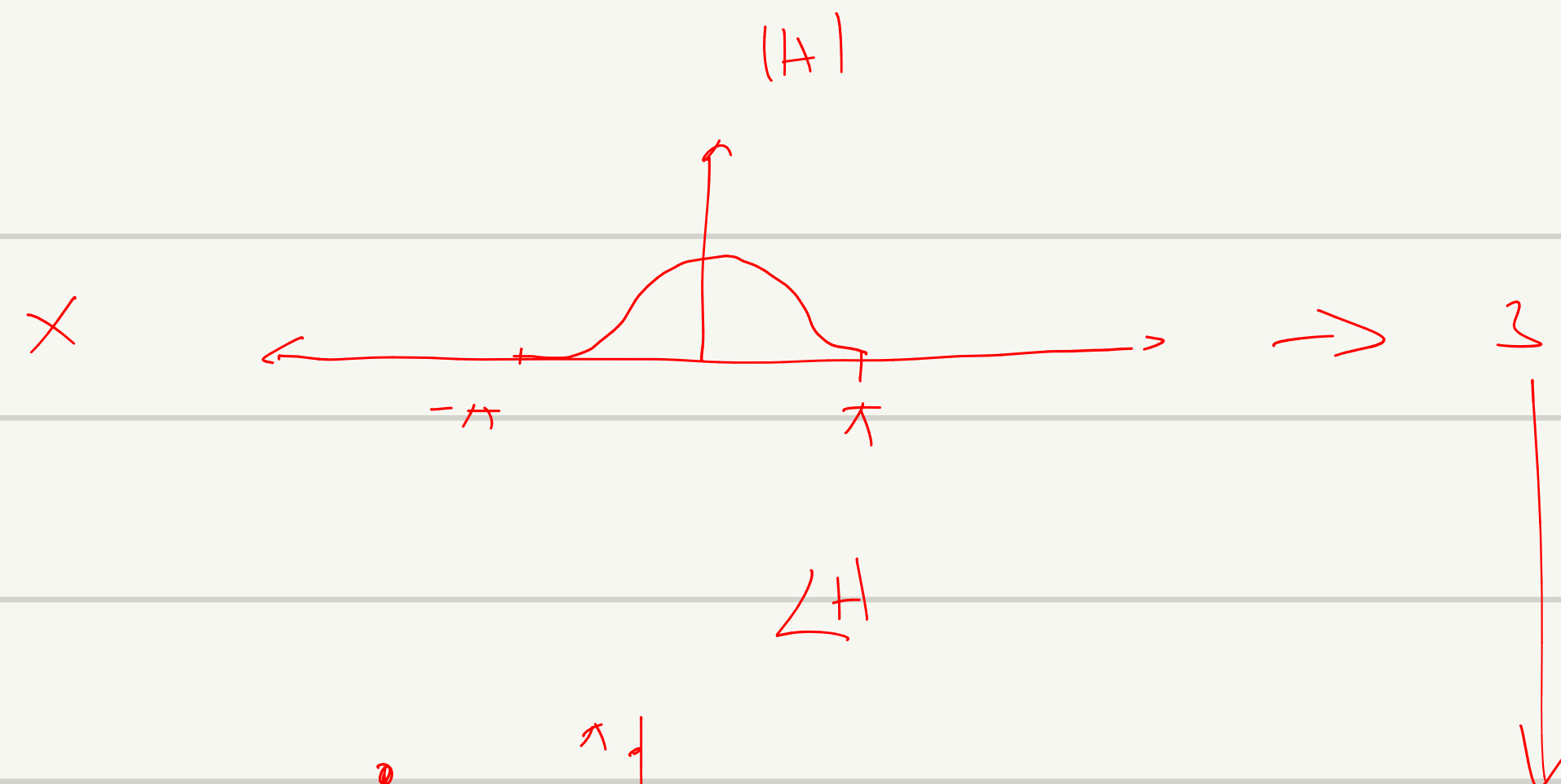
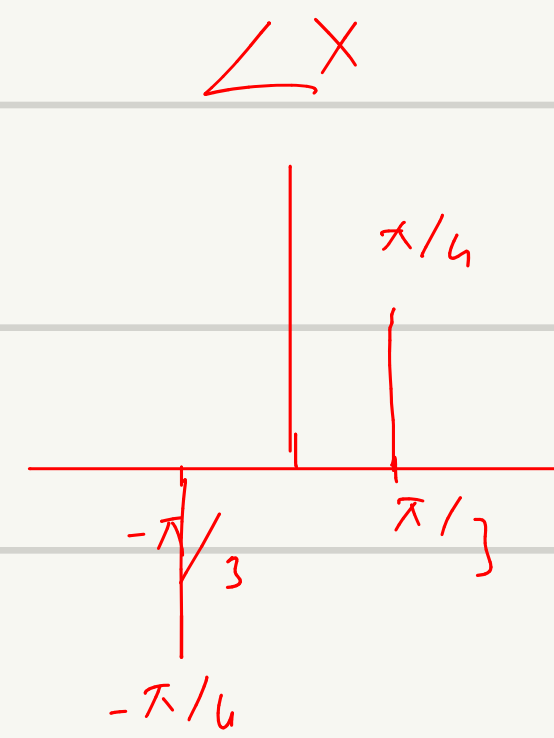
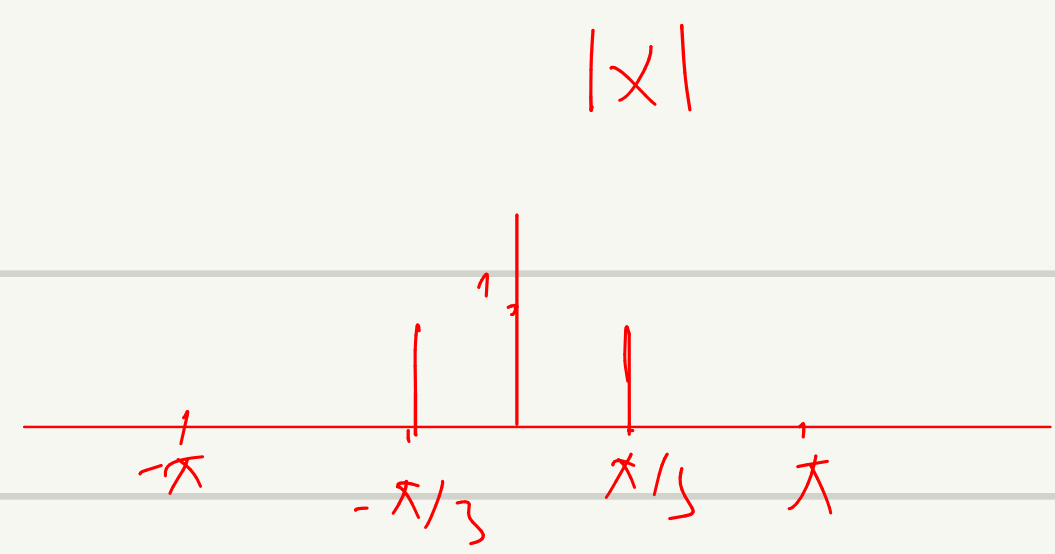
$$y_1[n] = H(e^{j\pi/3})e^{j(\pi n/3 + \pi/4)} = 3e^{-j(\pi/3)}e^{j(\pi n/3 + \pi/4)}$$

$$y_2[n] = H(e^{-j\pi/3})e^{-j(\pi n/3 + \pi/4)} = 3e^{j(\pi/3)}e^{-j(\pi n/3 + \pi/4)}$$

$$y[n] = 3e^{j(\pi n/3 - \pi/12)} + 3e^{-j(\pi n/3 - \pi/12)}$$

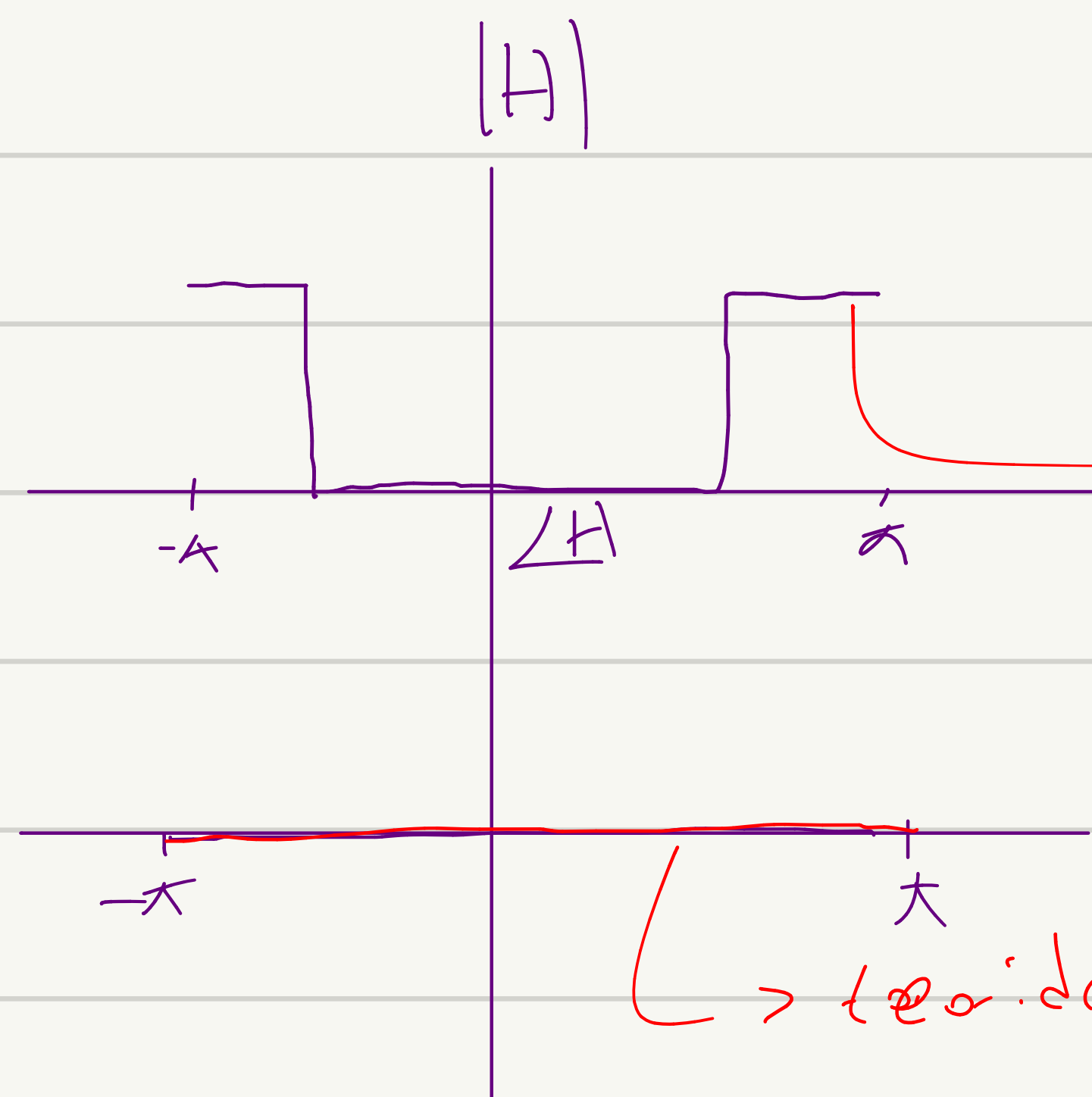
$$\Rightarrow y[n] = 6 \cos(\frac{\pi}{3}n - \frac{\pi}{12})$$





$$\rightarrow \left(-\frac{\pi}{6} + \frac{\pi}{3} \right) \quad \left(\frac{\pi}{6} - \frac{\pi}{3} \right)$$

$$\rightarrow \frac{\pi}{6} \quad -\frac{\pi}{6}$$



$$x[n] = 5 \cdot \cos(\pi n) + 3 \cdot \cos(2\pi n)$$

$$y[n] = 5 \cdot \cos(\pi n)$$

→ köşeli: yapılar oluşturmak için çok uzun H logoları gerekir

→ teoride böyle olur ama pratikte → böyle olmaz.

$$h[n] = f[n] - f[n-1] \rightarrow \sum_{k=0}^1 h[k] e^{-j\omega k} \rightarrow 1 - e^{-j\omega} = |H(e^{j\omega})|$$

$$\hookrightarrow 1 - \cos \hat{\omega} + j \sin \hat{\omega}$$

$$A = \sqrt{(1 - \cos \hat{\omega})^2 + (\sin \hat{\omega})^2}$$

$$\hat{\omega} = 0 \quad \text{in} \rightarrow \sqrt{(1-1)^2 + 0^2} = 0$$

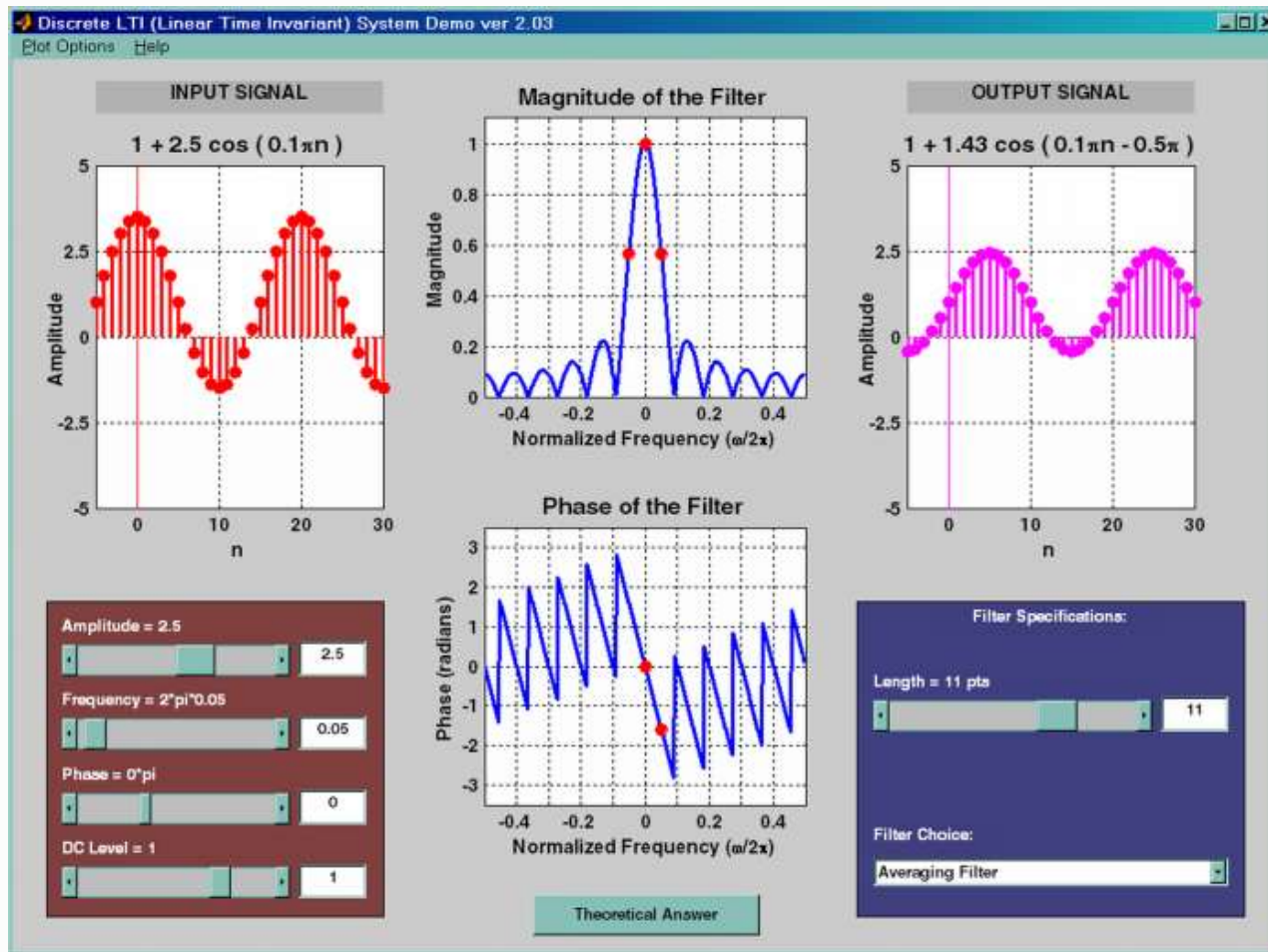
$$\hat{\omega} = \pi \quad \text{in} \rightarrow \sqrt{2^2 + 0^2} \rightarrow 2$$

$$\sum_{k=0}^1 h[k] \cdot e^{-j\omega k}$$

$$1 - e^{-j\omega}$$

$$|H|$$

DLTI Demo with Sinuzoids

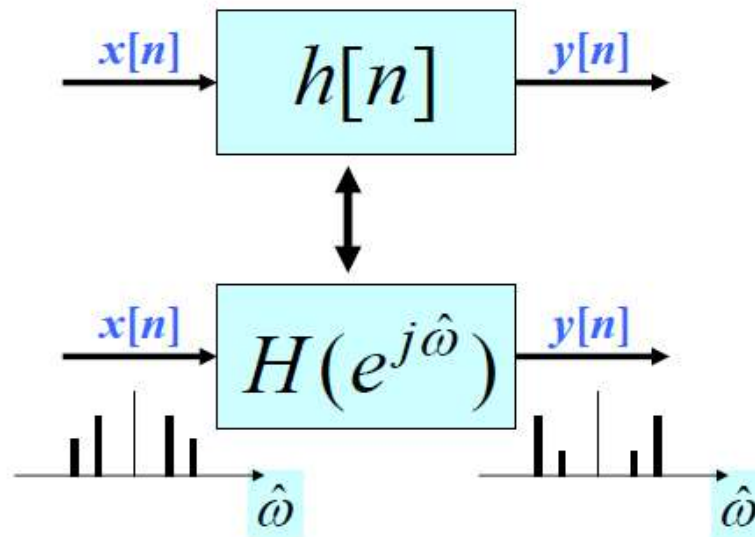


<https://dspfirst.gatech.edu/matlab/#dltidemo>

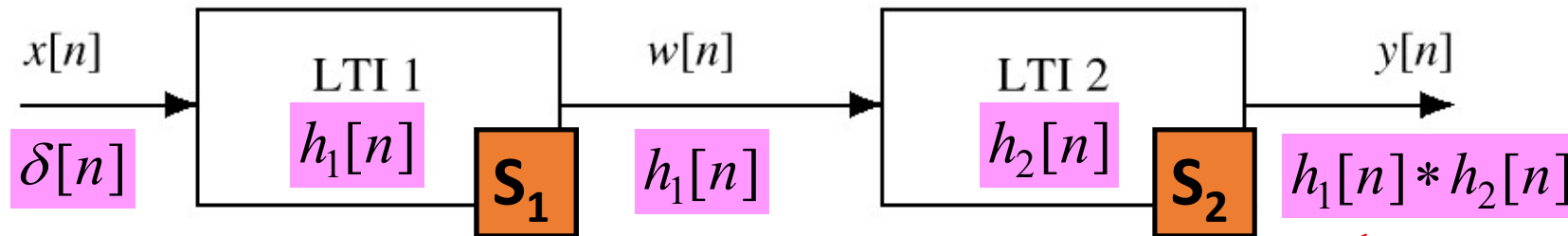
Summary over Block Diagrams



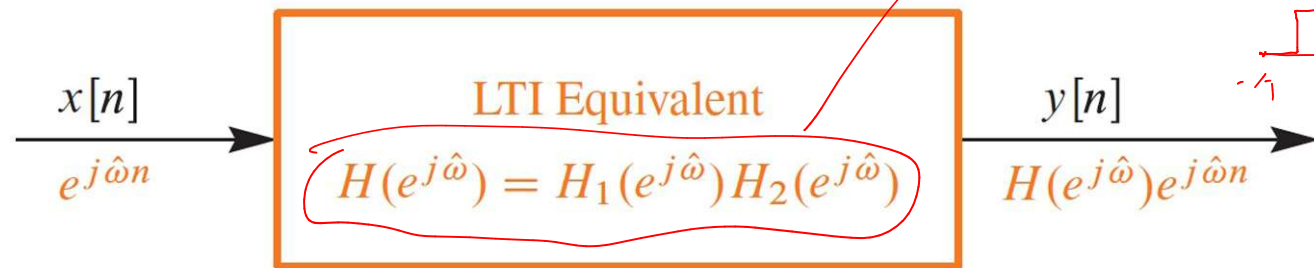
- Equivalent Representations



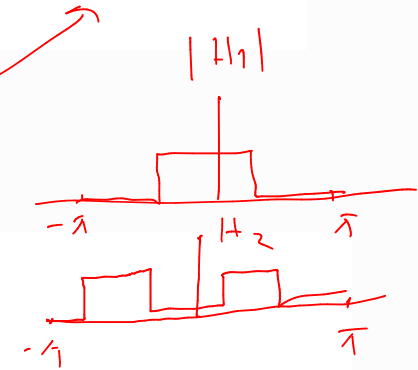
Cascaded LTI Systems



WHAT is the overall FREQUENCY RESPONSE ?



$$H(e^{j\hat{\omega}}) = H_1(e^{j\hat{\omega}})H_2(e^{j\hat{\omega}})$$



$H_1 \times H_2$

Band pass filter

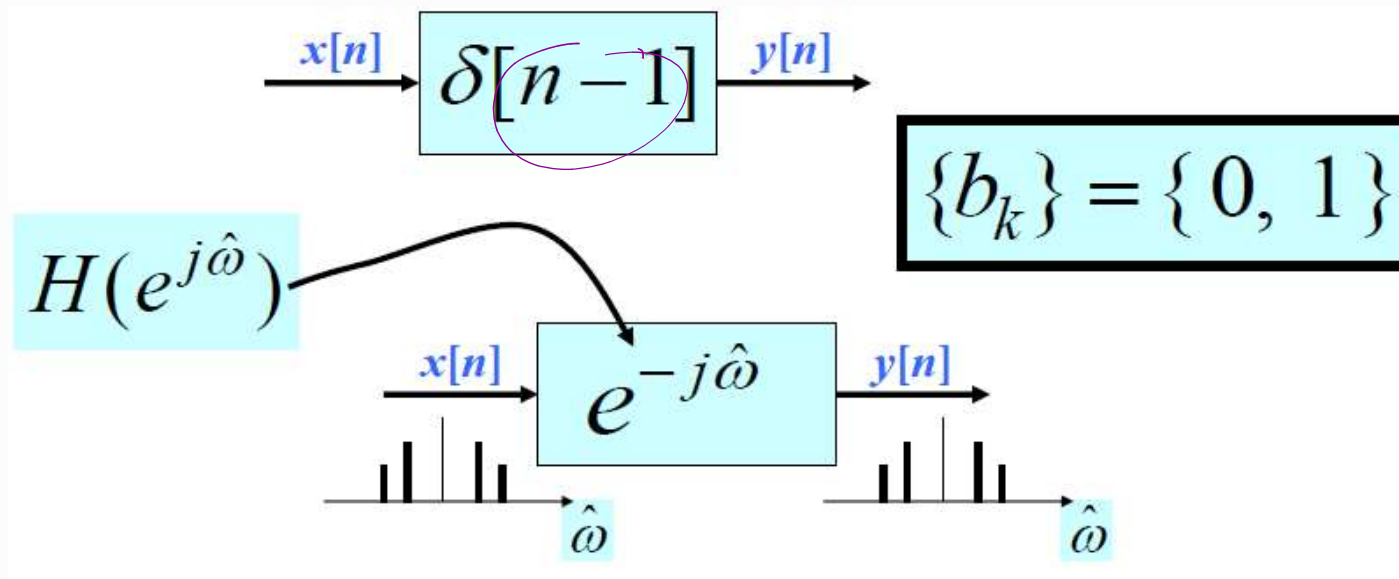


<https://dspfirst.gatech.edu/chapters/06firfreq/demos/blockd/index.html>

Example – 4: Unit Delay System

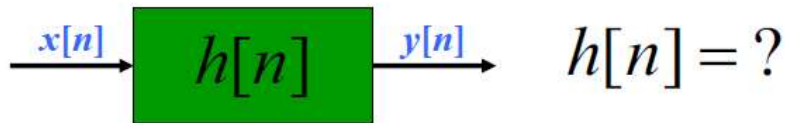
Find $h[n]$ and $H(e^{j\hat{\omega}})$ for $y[n] = x[n - 1]$

$$\sum_{k=-\infty}^{\infty} h[k] e^{-j\hat{\omega}k}$$



Example – 5: Freq. Domain to Time

$H(e^{j\hat{\omega}})$ and find $h[n]$ or b_k

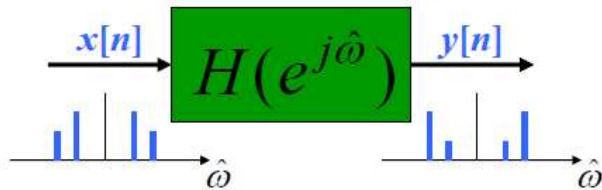


$$H(e^{j\hat{\omega}}) = 7e^{-j2\hat{\omega}} \cos(\hat{\omega})$$

$$H(e^{j\hat{\omega}}) = \sum_{k=0}^M b_k e^{-j\hat{\omega}k}$$

$$3.5 \frac{e^{j\omega} + e^{-j\omega}}{2}$$

$$(e^{-j\omega} + e^{-j3\omega})$$



$$b_1 = 3.5$$

$$b_3 = 3.5$$

$$h[n] = \{0, 3.5, 0, 3.5\}$$

$$H(e^{j\hat{\omega}}) = 7e^{-j2\hat{\omega}} \cos(\hat{\omega})$$

EULER's Formula

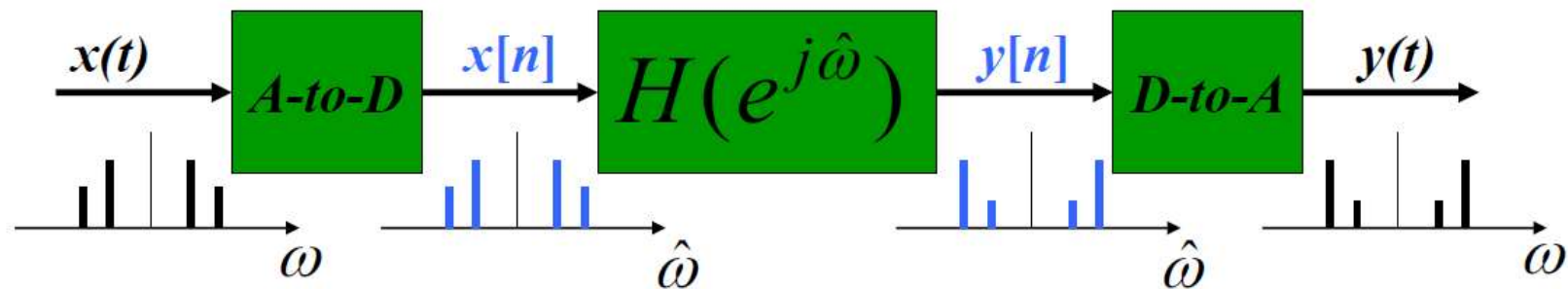
$$= 7e^{-j2\hat{\omega}} (0.5e^{j\hat{\omega}} + 0.5e^{-j\hat{\omega}})$$

$$= (3.5e^{-j\hat{\omega}} + 3.5e^{-j3\hat{\omega}})$$

$$h[n] = 3.5\delta[n-1] + 3.5\delta[n-3]$$

$$b_k = \{0, 3.5, 0, 3.5\}$$

Digital Filtering



ω – SPECTRUM of $x(t)$ (SUM of SINUSOIDS)

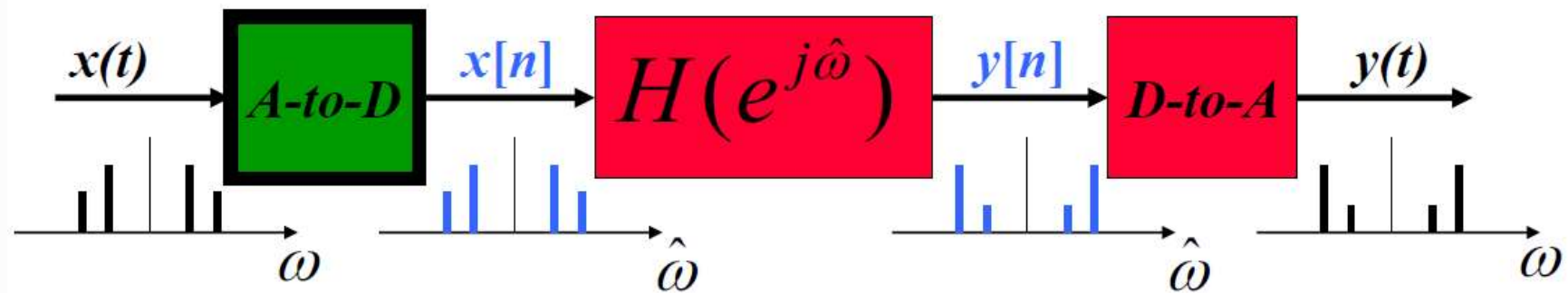
$\hat{\omega}$ – SPECTRUM of $x[n]$

- Is ALIASING a PROBLEM ?

$\hat{\omega}$ – SPECTRUM $y[n]$ (FIR Gain or Nulls)

ω – Then, OUTPUT $y(t)$ = SUM of SINUSOIDS

Frequency Scaling

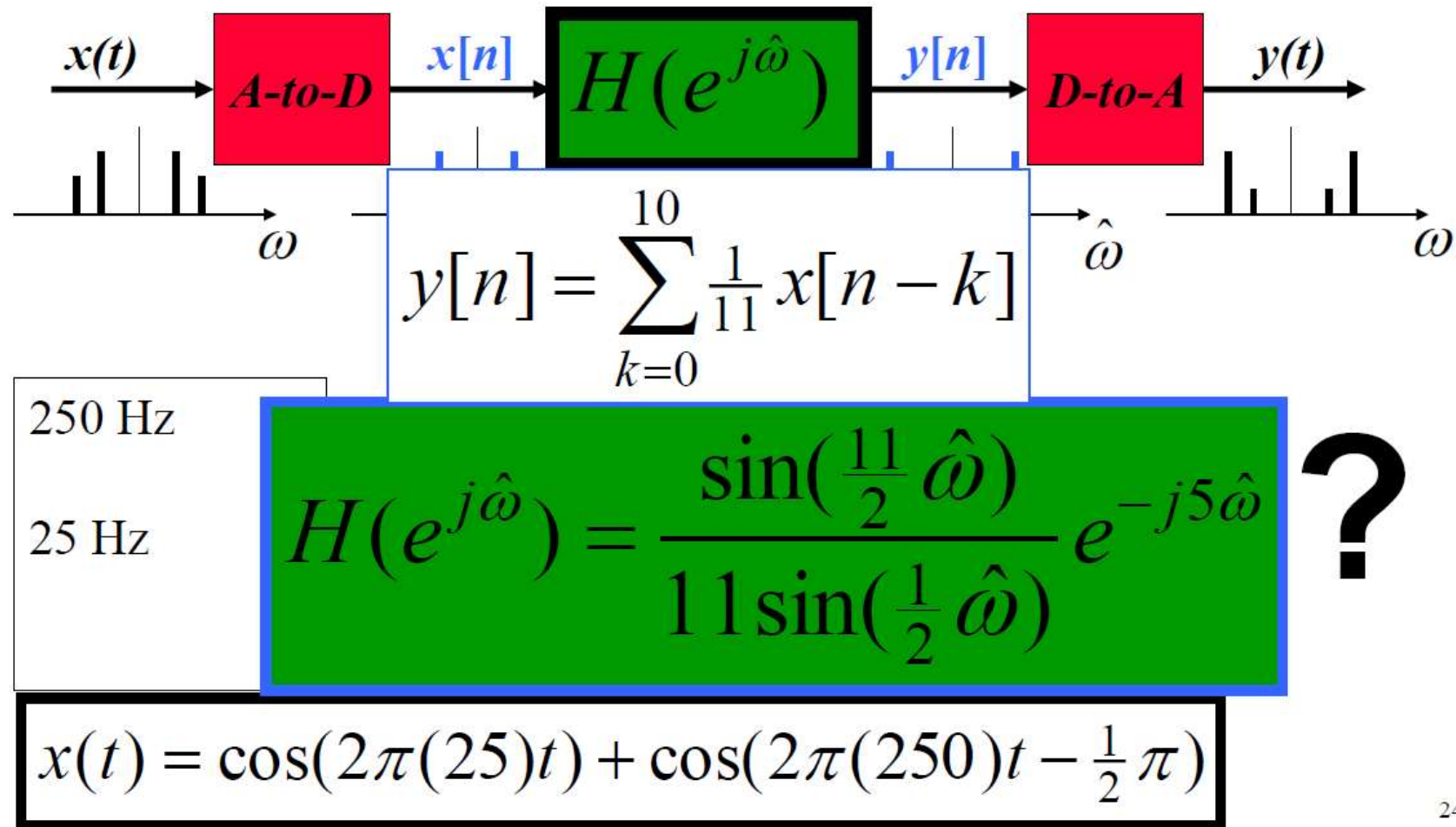


- TIME SAMPLING:
 - IF NO ALIASING:
 - FREQUENCY SCALING

$$t = nT_s$$

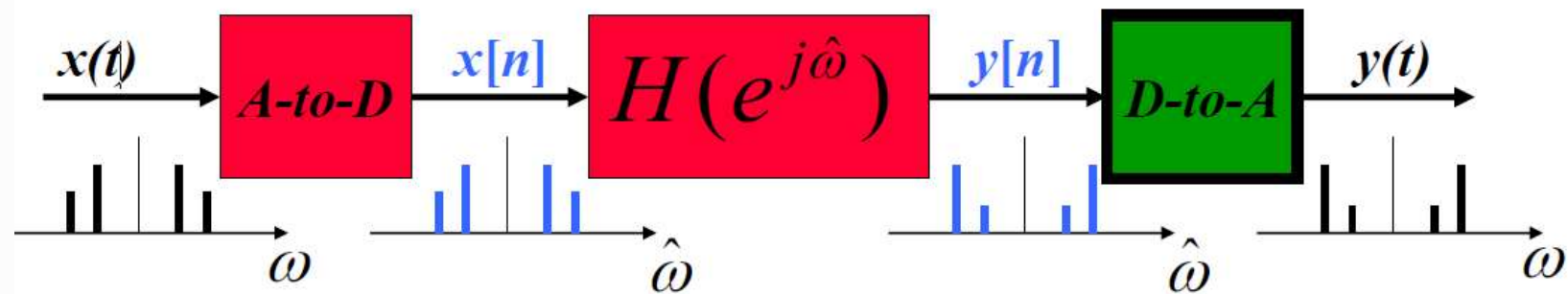
$$\hat{\omega} = \omega T_s = \frac{\omega}{f_s}$$

11-pt Averager



24

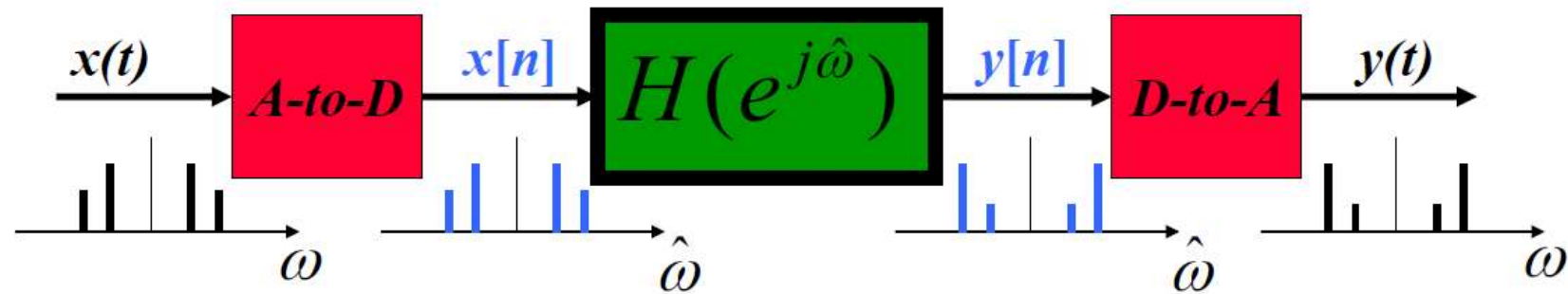
D-A Frequency Scaling



- TIME SAMPLING: $t = nT_s \Rightarrow n \leftarrow tf_s$
- RECONSTRUCT up to $0.5f_s$
– FREQUENCY SCALING

$$\omega = \hat{\omega} f_s$$

Summary



• 250 Hz	• 0.5π	$H(e^{j0.5\pi})$	• 0.5π	• 250 Hz
• 25 Hz	• $.05\pi$	$H(e^{j0.05\pi})$	• $.05\pi$	• 25 Hz

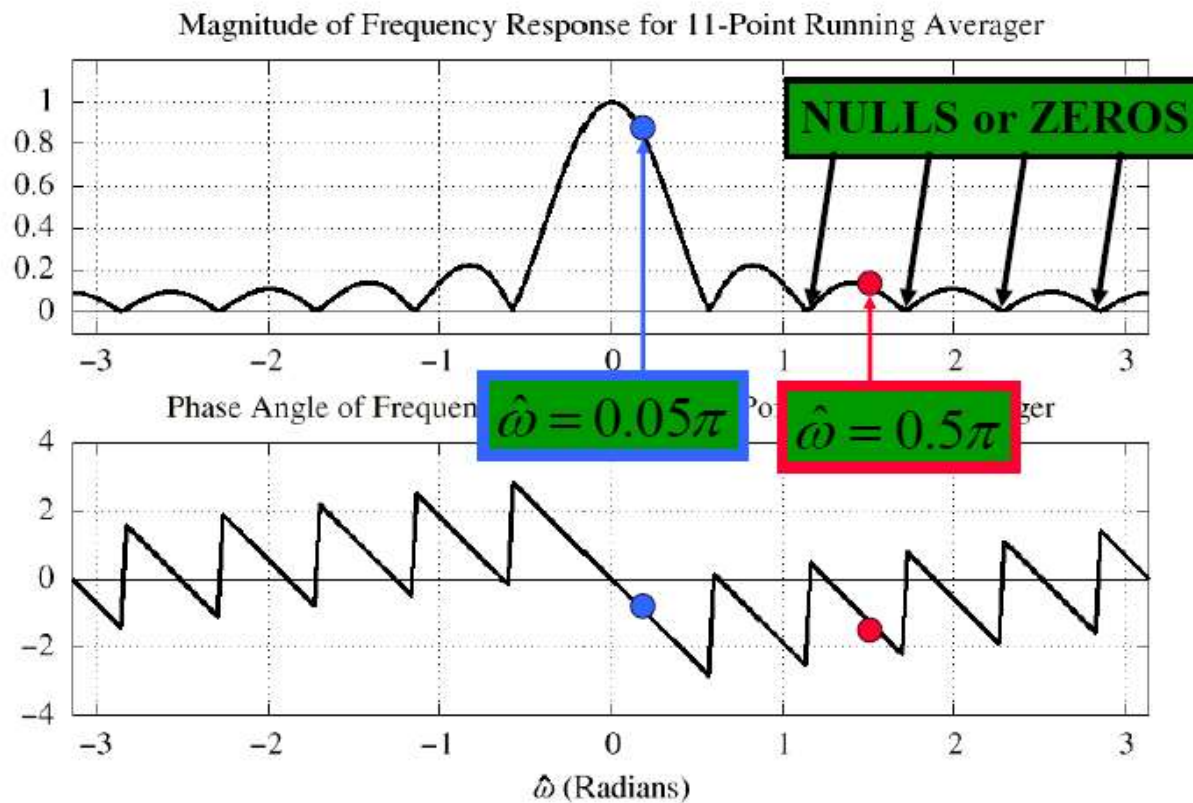
$F_s = 1000 \text{ Hz}$

NO new freqs

$$t = nTs = n/1000$$

$$\cos(2\pi 250t) \rightarrow \cos\left(2\pi \cdot 250 \cdot \frac{n}{1000}\right) = \cos(0.5\pi n)$$

Magnitude of Frequency Response



At $\hat{\omega} = 0.5\pi$

$$H(e^{j\hat{\omega}}) = \frac{\sin(\frac{11}{2}\hat{\omega})}{11\sin(\frac{1}{2}\hat{\omega})} e^{-j5\hat{\omega}}$$

$$H(e^{j\hat{\omega}}) = \frac{\sin(\frac{11}{2}(0.5\pi))}{11\sin(\frac{1}{2}(0.5\pi))} e^{-j5(0.5\pi)}$$

$$= \frac{\sin(2.75\pi)}{11\sin(0.25\pi)} e^{-j2.5\pi}$$

$$= 0.0909 e^{-j0.5\pi}$$

Magnitude of Frequency Response



$$\begin{aligned}
 H(e^{j2\pi(25)/1000}) &= \frac{\sin(\pi(25)(11)/1000)}{11 \sin(\pi(25)/1000)} e^{-j2\pi(25)(5)/1000} \\
 f_s &= 1000 \\
 H(e^{j2\pi(250)/1000}) &= \frac{\sin(\pi(250)(11)/1000)}{11 \sin(\pi(250)/1000)} e^{-j2\pi(250)(5)/1000} \\
 &= 0.8811 e^{-j\pi/4} \\
 &= 0.0909 e^{-j\pi/2} \\
 y(t) &= \underline{0.8811} \cos(2\pi(25)t - \underline{\pi/4}) + \underline{0.0909} \sin(2\pi(250)t - \underline{\pi/2})
 \end{aligned}$$

MAG SCALE

PHASE CHANGE

Remember: 17-pt Centralized Average filter to Noisy Audio

```
clc; clear all;
```

```
%% Load Sound
```

```
load ('piano2.mat');
```

```
x = x(1:16000);
```

```
soundsc(x, Fs);
```

```
%% Add noise
```

```
K = awgn(x, 40);
```

```
soundsc(K, Fs);
```

```
%% Filter
```

```
N = 17;
```

```
h = 1/N*ones(1, N);
```

```
%% Apply Convolution
```

```
y = conv(K, h, 'same');
```

```
soundsc(y, Fs);
```

```
%%
```

```
plot(x, 'r'); hold on; plot(y, 'b');
```

LOAD THE SIGNAL

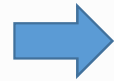
ADD A NOISE TO SIGNAL

FILTER THE SIGNAL

But How it works? What is the frequency response?

17-pt Averager

$$y(n) = \frac{1}{17} \sum_{k=0}^{16} x(n-k)$$

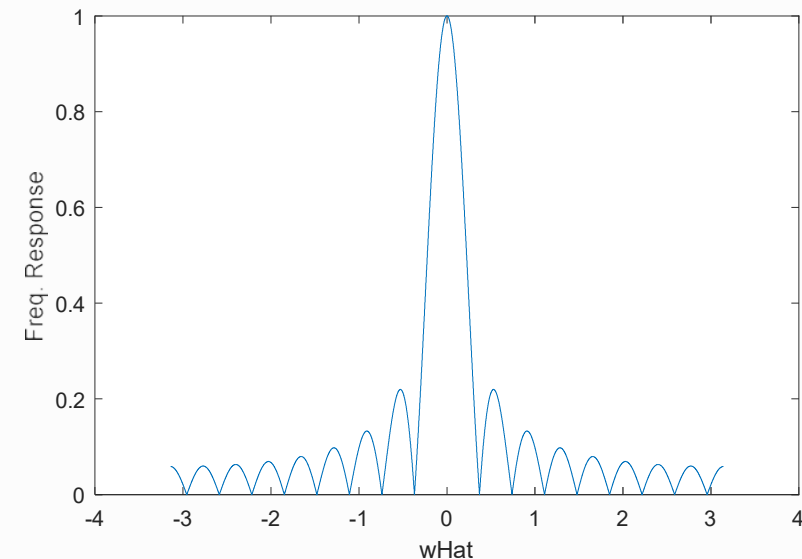


$$h(n) = \frac{1}{17} \sum_{k=0}^{16} \delta(n-k) = \frac{1}{17} \delta(n) + \dots + \frac{1}{17} \delta(n-16)$$

$$H(e^{j\hat{\omega}}) = \sum_{k=0}^M b_k e^{-j\hat{\omega}k}$$

```
wHat = linspace(-pi,pi,Fs);
b = (1/17)*ones(1,17);
%%
H = zeros(1,Fs);
for k = 1:17
    H = H + b(k)*exp(-1j*wHat*k);
end

plot (wHat, abs(H));
xlabel('wHat');
ylabel('Freq. Response');
```



Let's make a deep analysis

```

clc; clear all;
%% Load Sound
load ('piano2.mat');
x = x(1:16000);
X = fftshift(fft(x,Fs));
wHat = linspace(-pi,pi,Fs);
plot(wHat,abs(X));
xlabel('wHat');
ylabel('Freq. Response X');

%% Add Noise
Xnoise = awgn(x,40);
Xnoisef = fftshift(fft(Xnoise,Fs));
figure(2);
plot(wHat,abs(Xnoisef));
xlabel('wHat');
ylabel('Freq. Response Xnoise');

%% Filter
N = 17; h = 1/N*ones(1,N);
y = conv(Xnoise,h,'same');
yf = fftshift(fft(y,Fs));
figure(3);
plot(wHat,abs(yf));
xlabel('wHat');
ylabel('Freq. Response Y');

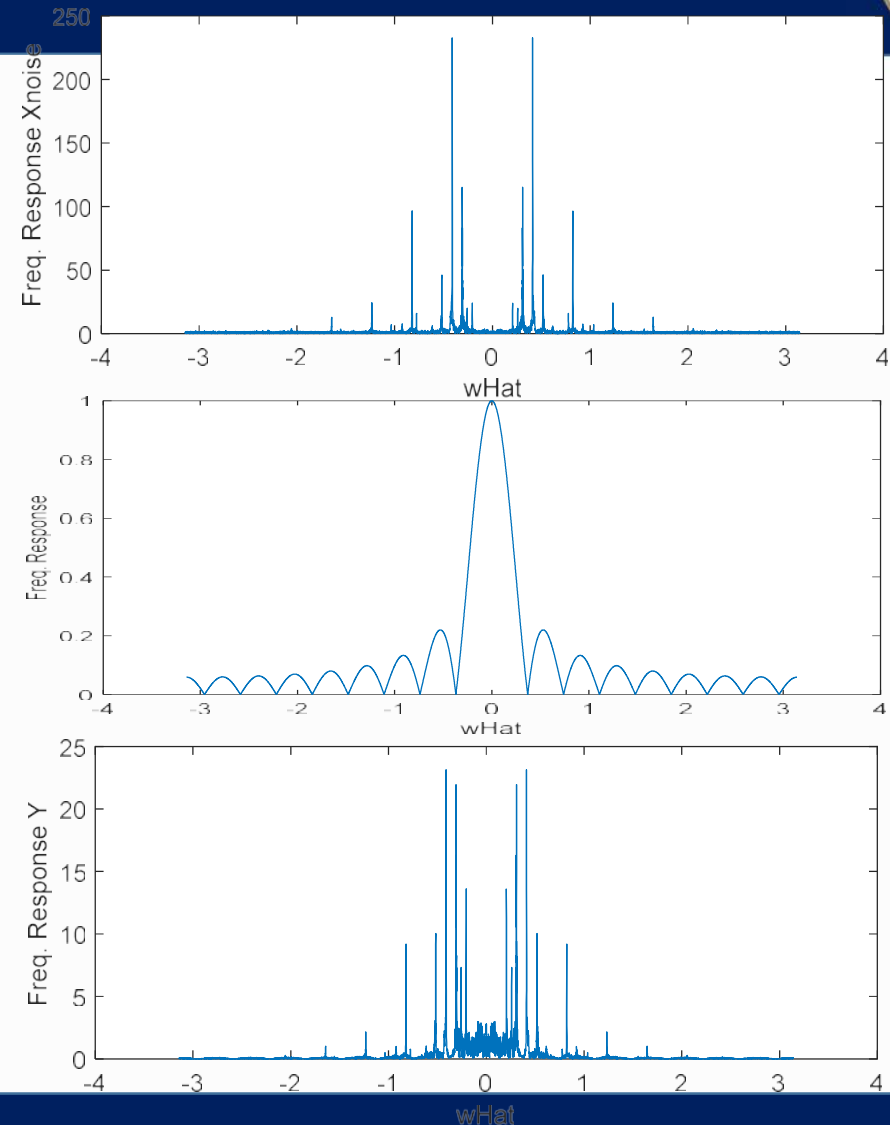
```

Input

X

Filter

Output



Hearing Test – Audiometry Test (Try it at home)

Conduct a test of your hearing, and present the results as a frequency response plot.

Define a sampling frequency (F_s)

From 20 Hz to 22000 Hz with 100 Hz step do:

Play a tone with the selected frequency

Did you hear it: Give a score from 0-100.

Save this value for the last plot

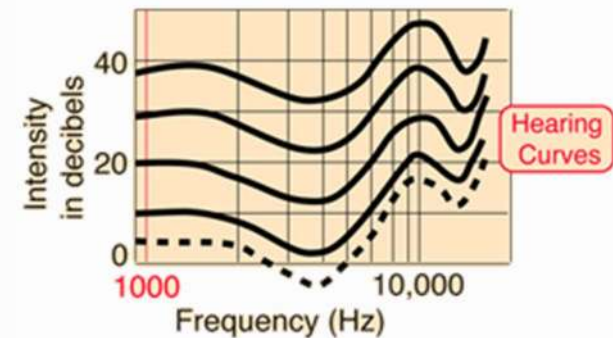
Continue loop.

- Plot analog frequency vs. $|H|$. (Freq in logspace)
- Plot digital frequency vs. $|H|$. (Freq in logspace)

<https://dspfirst.gatech.edu/chapters/06firfreq/labs/HearingTestFreqResponse/HearingTestFreqResponse.pdf>

<http://hyperphysics.phy-astr.gsu.edu/hbase/Sound/maxsens.html>

Use the hearing test to determine the frequency where your hearing sensitivity starts to drop significantly.



Exercise - 1



A linear time-invariant system is described by the FIR difference equation

$$y[n] = x[n] - 3x[n - 1] + 9x[n - 2] - 3x[n - 3] + x[n - 4]$$

(a) Write a simple formula for the magnitude of the frequency response $|H(e^{j\hat{\omega}})|$. Express your answer in terms of real-valued functions only.

(b) Derive a simple formula for the phase of the frequency response $\angle H(e^{j\hat{\omega}})$.

$$\begin{aligned} H(e^{j\hat{\omega}}) &= 1 - 3e^{-j\hat{\omega}} + 9e^{-j2\hat{\omega}} - 3e^{-j3\hat{\omega}} + e^{-j4\hat{\omega}} \\ &= e^{-j2\hat{\omega}} (e^{j2\hat{\omega}} - 3e^{j\hat{\omega}} + 9 - 3e^{-j\hat{\omega}} + e^{-j2\hat{\omega}}) \\ &= e^{-j2\hat{\omega}} (2\cos 2\hat{\omega} - 6\cos \hat{\omega} + 9) \end{aligned}$$

$\Rightarrow$

$$|H(e^{j\hat{\omega}})| = 2\cos 2\hat{\omega} - 6\cos \hat{\omega} + 9$$

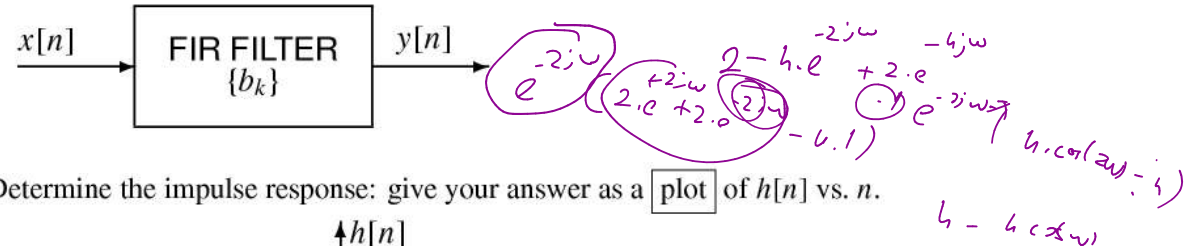
$$\begin{aligned} &1 - 3e^{-j\hat{\omega}} + 9e^{-j2\hat{\omega}} - 3e^{-j3\hat{\omega}} + e^{-j4\hat{\omega}} \\ &= e^{-j2\hat{\omega}} (e^{j2\hat{\omega}} - 3e^{j\hat{\omega}} + 9 - 3e^{-j\hat{\omega}} + e^{-j2\hat{\omega}}) \\ &= e^{-j2\hat{\omega}} (2\cos(2\hat{\omega}) - 6\cos(\hat{\omega}) + 9) \\ &\angle H(e^{j\hat{\omega}}) = -2\hat{\omega} \end{aligned}$$

Exercise - 2

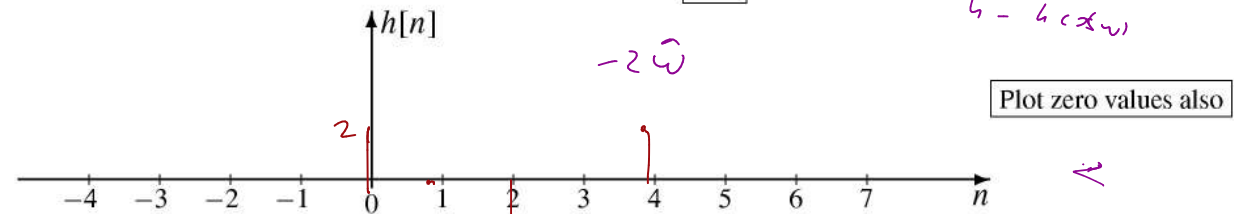


PROBLEM:

The following FIR filter is specified by the filter coefficients $\{b_k\} = \{2, 0, -4, 0, 2\}$



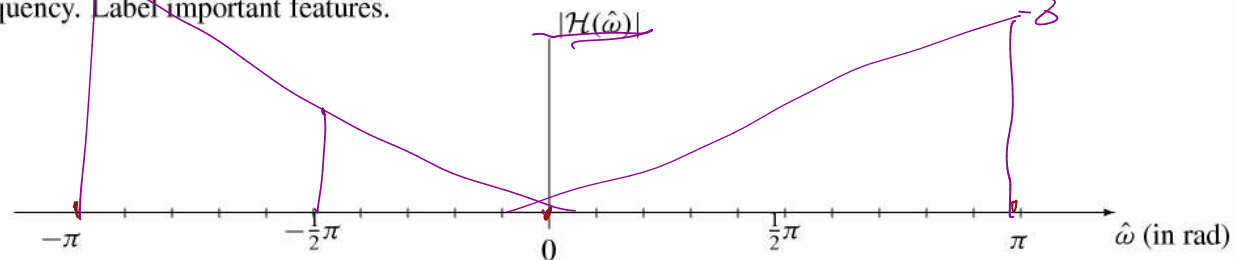
- (a) Determine the impulse response: give your answer as a **plot** of $h[n]$ vs. n .



- (b) Determine the frequency response, $\mathcal{H}(\hat{\omega})$, and select one of the following as the correct answer:

(A) $(4 - 4 \cos(2\hat{\omega}))e^{-j(2\hat{\omega}-\pi)}$ (B) $2 \cos \hat{\omega} + 4e^{-j(2\hat{\omega}+\pi)}$ (C) $(4 \cos(2\hat{\omega}) - 4)e^{-j\hat{\omega}}$
(D) $2 \cos(2\hat{\omega}) - 4$

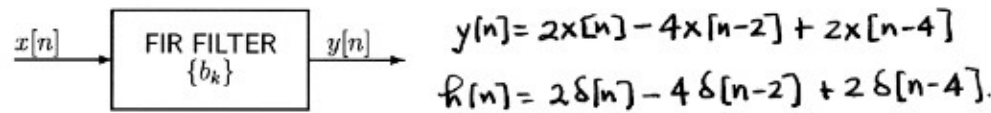
- (c) Determine the magnitude of $\mathcal{H}(\hat{\omega})$ and present your answer as a **plot** of the magnitude vs. frequency. Label important features.



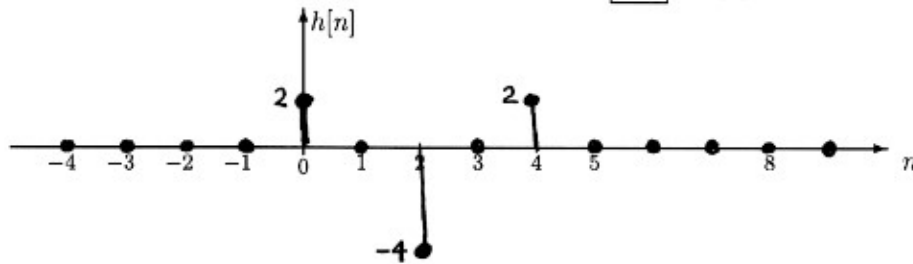
Solution



The following FIR filter is specified by the filter coefficients $\{b_k\} = \{2, 0, -4, 0, 2\}$



(a) Determine the impulse response: give your answer as a of $h[n]$ vs. n .



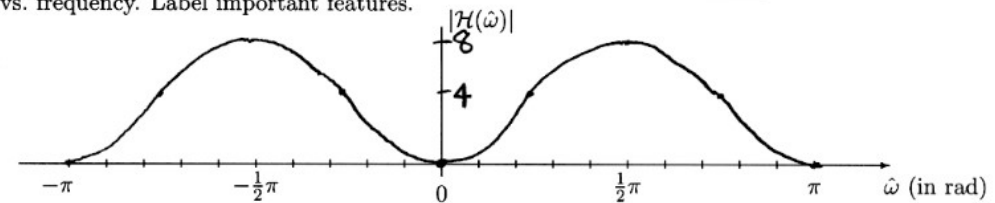
(b) Determine the frequency response, $\mathcal{H}(\hat{\omega})$, and select one of the following as the correct answer:

- ☒ (A) $(4 - 4\cos(2\hat{\omega}))e^{-j(2\hat{\omega}-\pi)}$ (B) $2\cos\hat{\omega} + 4e^{-j(2\hat{\omega}+\pi)}$ (C) $(4\cos(2\hat{\omega}) - 4)e^{-j\hat{\omega}}$
 (D) $2\cos(2\hat{\omega}) - 4$

$$\begin{aligned}
 \mathcal{H}(\hat{\omega}) &= 2 - 4e^{-j2\hat{\omega}} + 2e^{-j4\hat{\omega}} \\
 &= e^{-j2\hat{\omega}}(2e^{+j2\hat{\omega}} - 4 + 2e^{-j2\hat{\omega}}) \\
 &= e^{-j2\hat{\omega}}(4\cos 2\hat{\omega} - 4) \\
 &= e^{-j2\hat{\omega}}e^{j\pi}(4 - 4\cos 2\hat{\omega})
 \end{aligned}$$

(A)

(c) Determine the magnitude of $\mathcal{H}(\hat{\omega})$ and present your answer as a of the magnitude vs. frequency. Label important features.



$$\begin{aligned}
 |\mathcal{H}(\hat{\omega})| &= |4 - 4\cos 2\hat{\omega}| = 4 - 4\cos 2\hat{\omega} \\
 &\text{This is non-negative}
 \end{aligned}$$

$\begin{aligned}
 \hat{\omega} = 0 &\Rightarrow 4 - 4 = 0 \\
 \hat{\omega} = \pi &\Rightarrow 4 - 4 = 0 \\
 \hat{\omega} = \pi/2 &\Rightarrow 4 - 4(-1) = 8 \\
 \hat{\omega} = \pi/4 &\Rightarrow 4 - 4(0) = 4
 \end{aligned}$

Exercise - 3



PROBLEM:

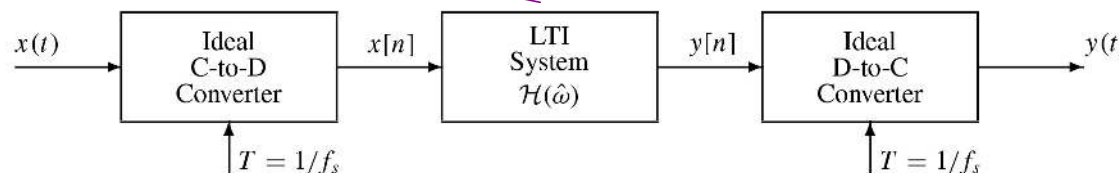
The input to the C-to-D converter in the figure below is

$$x(t) = 3 + 2 \cos(6000\pi t - \pi/4) + 11 \cos(12000\pi t - \pi/3)$$

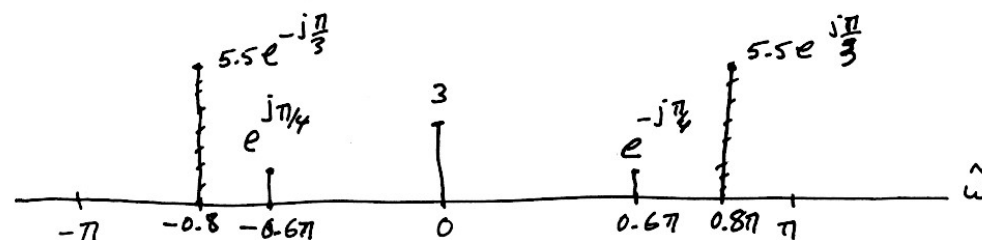
The frequency response for the digital filter (LTI system) is

$$H(\hat{\omega}) = \frac{\sin(5\hat{\omega})}{\sin(\frac{1}{2}\hat{\omega})} e^{-j5\hat{\omega}}$$

If $f_s = 10000$ samples/second, determine an expression for $y(t)$, the output of the D-to-C converter.



$$x[n] = 3 + 2 \cos(0.6\pi n - \pi/4) + 11 \cos(1.2\pi n - \pi/3)$$



$$H(\hat{\omega}) = \frac{\sin(5\hat{\omega})}{\sin(\frac{1}{2}\hat{\omega})} e^{-j5\hat{\omega}}$$

$$H(0) = 10$$

$$H(0.6\pi) = \frac{\sin 3\pi}{\sin 0.3\pi} e^{-j3\pi} = 0$$

$$H(0.8\pi) = \frac{\sin 4\pi}{\sin 0.4\pi} e^{-j4\pi} = 0$$

$$y[n] = 10 \times 3 = 30, \quad y(t) = 30$$